# Supplementary File

## Appendix I: Zoster Vaccination

| Country | Recommendation | Vaccine type |
|---|---|---|
| New Zealand | Adults aged 65 years,<br>catch-up 66–80 years (end by 31 December 2020) | ZVL (Zostavax®) |
| Australia | Adults aged 70 years, and<br>Catch-up 71–79 years (five-year program) until 2021 | ZVL (Zostavax®) |
| USA | Adults age 50 and older | RZV (Shingrix®) |
| Austria | Adults aged ≥50 years | RZV (Shingrix®) |
| Germany | Adults aged ≥60 years and specific groups | RVZ (Shingrix®) |
| Canada | Adults aged ≥50 years | RVZ (Shingrix®) |
| France | Adults aged 65–74 years,<br>Catch-up 75–79 | ZVL (Zostavax®) |
| Spain | Adults aged ≥60 years (for specific risk groups) | ZVL (Zostavax®) |
| Greece | Adults aged ≥60 years | ZVL (Zostavax®) |
| Italy | Adults aged ≥65 years | ZVL (Zostavax®) |
| United Kingdom | Adults aged 70–78 years | ZVL (Zostavax®) |
| United Arab Emirates | Adults aged >60 years | ZVL (Zostavax®) |

## Appendix II: Outcome Definitions

1. Confirmed herpes zoster infection: Those who underwent clinical diagnosis of herpes zoster infection through laboratory testing (e.g. polymerase-chain-reaction assay, virus culture) and/or examination by a physician were classified as having confirmed cases of herpes zoster infection (1).

2. Suspected herpes zoster infection: Subjects with non-injection-site herpes-zoster like rashes, varicella-like rashes, vesicular rashes or unilateral rashes (as defined by the investigators) and no other alternative clinical diagnoses were classified as having suspected cases of herpes zoster infection (1).

3. Herpes zoster ophthalmicus: Subjects with herpes zoster infection in the ocular region (1). It is the reactivation of latent varicella-zoster virus (VZV) in the first and/or second division of the trigeminal nerve (2)

4. Post-herpetic neuralgia: Subjects with pain continuing 90 days or longer after the onset of the shingles rash. Effectiveness data are based on this definition and also known as PHN-90

(1). Any pain persisting after rash healing to pain that has persisted at least 60, 90, 120 days or 6 months after rash onset to clinically meaningful pain 3 months or more after rash onset (2).

5. Quality-of-life: Measured by EuroQol (EQ5D), Health Utilities Index Mark 2 (HUI2), Health Utilities Index Mark 3 (HUI3) or Short Form 6 Dimensions (SF6D) (1)

References

1. Tricco AC, Zarin W, Cardoso R, Veroniki A-A, Khan PA, Nincic V, et al. Efficacy, effectiveness, and safety of herpes zoster vaccines in adults aged 50 and older: systematic review and network meta-analysis. BMJ. 2018;363:k4029.
2. Watson CPN, Gershon AA, Oxman MN. Herpes zoster: postherpetic neuralgia and other complications: Springer; 2017.

**Appendix III: Search strategy**

Key concepts (alternative words)

1. Herpes zoster (zoster, shingles, varicella-zoster virus, human herpesvirus 3)

2. Postherpetic neuralgia (PHN)

3. Herpes zoster ophthalmicus (HZO)

4. Herpes zoster vaccine (zoster vaccine, shingles vaccine, zostavax, live attenuated zoster vaccine, zoster vaccine live (ZVL), shingrix, adjuvant recombinant subunit vaccine (RZV), varicella-zoster vaccine).

5. Effectiveness (effective, performance, impact, efficacy, effect, efficiency).

6. Other search terms to be considered: HZ related hospitalisations, quality of life, immune status, comorbidities.

Search conducted on Medline (Ovid) show the following

| # | Search term | Results |
|---|---|---|
| 1 | exp Herpes Zoster/ | 11747 |
| 2 | Herpesvirus 3, Human/ | 7028 |
| 3 | Varicellovirus/ | 285 |
| 4 | shingles.tw. | 1240 |
| 5 | (varicella adj3 virus*).tw. | 8299 |
| 6 | zoster.tw. | 17343 |
| 7 | (hhv3 or hhv-3).tw. | 16 |
| 8 | shingles.ab,ti. | 1240 |
| 9 | zoster.ab,ti. | 17343 |
| 10 | (varicella adj4 virus*).ab,ti. | 8370 |
| 11 | varicellovirus/de | 27 |
| 12 | varicellovir*.ab,ti. | 62 |
| 13 | hhv3.ab,ti. | 5 |
| 14 | 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 | 22283 |
| 15 | exp Vaccines/ | 225746 |
| 16 | exp Immunization/ | 173130 |
| 17 | Vaccination/ | 80276 |
| 18 | (vaccin* or immuni* or inocul*).tw. | 636177 |
| 19 | (vaccin* or immuni* or inocul*).ab,ti. | 636177 |
| 20 | 15 or 16 or 17 or 18 or 19 | 724546 |
| 21 | 14 and 21 | 4827 |
| 22 | Herpes Zoster Vaccine/ | 790 |
| 23 | ((zoster or shingles) adj3 vaccin*).tw. | 1264 |
| 24 | zostavax.tw,nm. | 132 |
| 25 | ((sub-unit or subunit) adj3 (vaccin* or zoster)).ab,ti. | 5893 |
| 26 | (Zostavax or Shingrix).ab,ti. | 153 |
| 27 | 22 or 23 or 24 or 25 or 26 | 7342 |
| 28 | 21 or 27 | 10650 |
| 29 | exp cohort analysis/ | 1978236 |
| 30 | exp longitudinal study/ | 132972 |
| 31 | exp prospective study/ | 534668 |
| 32 | cohort*.tw. | 573867 |
| 33 | exp case control study/ | 1069768 |
| 34 | (case* and control*).tw. | 688836 |
| 35 | 29 or 30 or 31 or 32 or 33 or 34 | 2751394 |
| 36 | 28 and 35 | 989 |
| 37 | limit 36 to yr="2006 -Current" | 696 |
| 38 | remove duplicates from 37 | 692 |

## Appendix IV: The JBI Critical Appraisal tools for use in JBI Systematic Reviews

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### JBI Critical Appraisal Checklist for Cohort Studies

Reviewer_____Date_____

Author _____Year_____Record Number_____.

| | Yes | No | Unclear | Not applicable |
|---|---|---|---|---|
| 1. Were the two groups similar and recruited from the same population? | ☐ | ☐ | ☐ | ☐ |
| 2. Were the exposures measured similarly to assign people to both exposed and unexposed groups? | ☐ | ☐ | ☐ | ☐ |
| 3. Was the exposure measured in a valid and reliable way? | ☐ | ☐ | ☐ | ☐ |
| 4. Were confounding factors identified? | ☐ | ☐ | ☐ | ☐ |
| 5. Were strategies to deal with confounding factors stated? | ☐ | ☐ | ☐ | ☐ |
| 6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)? | ☐ | ☐ | ☐ | ☐ |
| 7. Were the outcomes measured in a valid and reliable way? | ☐ | ☐ | ☐ | ☐ |
| 8. Was the follow up time reported and sufficient to be long enough for outcomes to occur? | ☐ | ☐ | ☐ | ☐ |
| 9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored? | ☐ | ☐ | ☐ | ☐ |
| 10. Were strategies to address incomplete follow up utilized? | ☐ | ☐ | ☐ | ☐ |
| 11. Was appropriate statistical analysis used? | ☐ | ☐ | ☐ | ☐ |

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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## JBI Critical Appraisal Checklist for Case Control Studies

Reviewer_____Date_____

Author _____Year_____Record Number_____

| | Yes | No | Unclear | Not applicable |
|---|---|---|---|---|
| 1. Were the groups comparable other than the presence of disease in cases or the absence of disease in controls? | ☐ | ☐ | ☐ | ☐ |
| 2. Were cases and controls matched appropriately? | ☐ | ☐ | ☐ | ☐ |
| 3. Were the same criteria used for identification of cases and controls? | ☐ | ☐ | ☐ | ☐ |
| 4. Was exposure measured in a standard, valid and reliable way? | ☐ | ☐ | ☐ | ☐ |
| 5. Was exposure measured in the same way for cases and controls? | ☐ | ☐ | ☐ | ☐ |
| 6. Were confounding factors identified? | ☐ | ☐ | ☐ | ☐ |
| 7. Were strategies to deal with confounding factors stated? | ☐ | ☐ | ☐ | ☐ |
| 8. Were outcomes assessed in a standard, valid and reliable way for cases and controls? | ☐ | ☐ | ☐ | ☐ |
| 9. Was the exposure period of interest long enough to be meaningful? | ☐ | ☐ | ☐ | ☐ |
| 10. Was appropriate statistical analysis used? | ☐ | ☐ | ☐ | ☐ |

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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